Experiment No.	02
Experiment Title	Email Spam/Ham Classification using Naive Bayes and KNN
GitHub Repository:	https://github.com/KJayasuriya/ml_laboratory
Student Name	K. Jayasuriya
Register Number	3122247001026
Faculty	Dr. Poreddy Ajaykumar Reddy
Submission Date	July 30, 2026

1 Objective

- Build Spam classifier using Naive Bayes and KNN.
- Compare Gaussian, Multinomial and Bernoulli Naive Bayes.
- Study the effect of varying k.
- Compare KDTree and BallTree.
- Optimize KNN using GridSearchCV and RandomizedSearchCV.
- Evaluate computational complexity and execution time.
- Evaluate using K-Fold Validation.

2 Problem Statement

The objective of this experiment is to develop an Email Spam/Ham Classification model using supervised machine learning algorithms. The experiment compares the performance of three variants of Naive Bayes classifiers (Gaussian, Multinomial and Bernoulli) with the K-Nearest Neighbors (KNN) algorithm. Hyperparameter tuning is performed using GridSearchCV and RandomizedSearchCV to identify the optimal KNN configuration. The performance of KDTree and BallTree search algorithms is also analyzed along with computational complexity, execution time and K-Fold Cross Validation.

3 Dataset Description

Dataset Name	Spambase
Dataset Source	https://www.kaggle.com/datasets/somesh24/spambase
Number of Samples	4601
Number of Features	58
Number of Classes	2
Missing Values	0
Train-Test Split	Train: 3680 and Test: 921

4 Exploratory Data Analysis

- Loaded the Spambase dataset and examined its dimensions and data types.
- Generated descriptive statistics using the `describe()` function.
- Verified that the dataset contains no missing values.
- Visualized the distribution of spam and ham emails.
- Analyzed correlations among numerical features using a heatmap.
- Examined feature distributions using histograms and boxplots to identify data spread and potential outliers.

Class Distribution

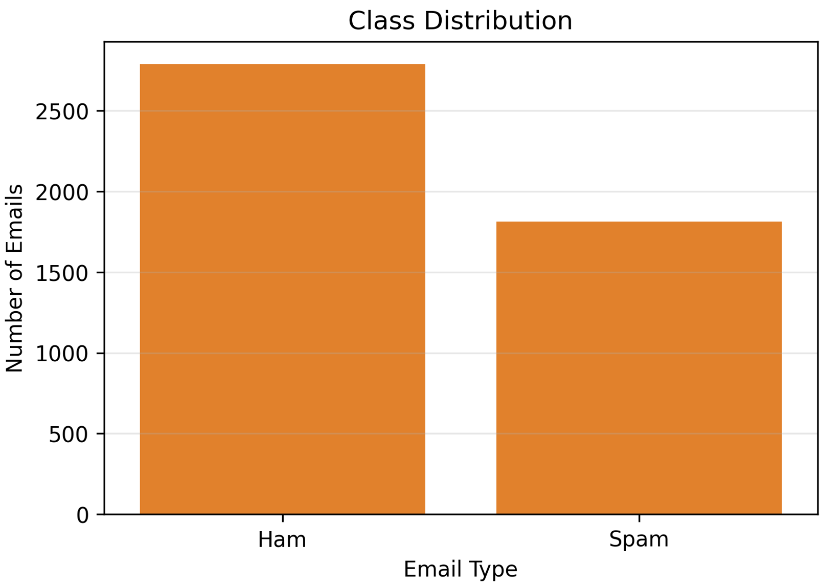

Figure 1: Class Distribution of Emails

Inference:
The class distribution plot shows that the dataset contains more ham emails than spam emails. Approximately 60.6% of the samples belong to the ham class, while 39.4% belong to the spam class. Although the dataset is not perfectly balanced, the class imbalance is moderate and suitable for training classification models without extensive resampling.

Correlation Matrix

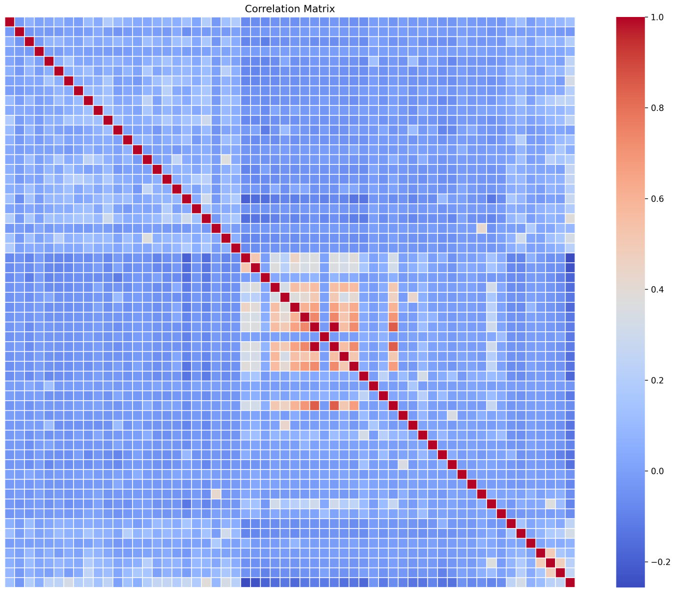

Figure 2: Correlation Matrix of Features

Inference:

The correlation matrix illustrates the relationships among the numerical features in the Spambase dataset. Most features exhibit weak to moderate correlations, while a few groups of features show stronger positive correlations. This indicates that the dataset contains diverse information with limited redundancy, making it suitable for spam classification.

5 Data Preprocessing

- **Missing Value Handling:**
  The Spambase dataset contains no missing values; therefore, no imputation or removal of records was required.
- **Encoding:**
  The target variable is already represented as binary labels (Spam/Ham), so no additional encoding was necessary.
- **Feature Scaling:**
  All numerical features were standardized using `StandardScaler`. Feature scaling is essential for KNN because it relies on distance calculations.
- **Train-Test Split:**
  The dataset was divided into training and testing sets using an 80:20 ratio, resulting in 3680 training samples and 921 testing samples.
- **Feature Engineering:**
  No additional feature engineering techniques were applied since the dataset already contains extracted numerical features.

6 Machine Learning Algorithm

- **Working Principle:**
  - **Naive Bayes:** Naive Bayes is a probabilistic supervised learning algorithm based on Bayes' theorem. It assumes that all features are conditionally independent given the class label and predicts the class with the highest posterior probability.
  - **K-Nearest Neighbors (KNN):** KNN is a non-parametric supervised learning algorithm that classifies a sample based on the majority class among its k nearest neighbors using a distance metric such as Euclidean or Manhattan distance.
- **Mathematical Formulation:**

  Bayes' theorem is given by

  $$P(C|X) = \frac{P(X|C) \times P(C)}{P(X)}$$

  where $P(C|X)$ is the posterior probability, $P(X|C)$ is the likelihood, $P(C)$ is the prior probability and $P(X)$ is the evidence.

  The Manhattan distance used in KNN is

  $$d(x, y) = \sum_{i=1}^{n} |x_i - y_i|$$

  where x and y are two data points.
- **Advantages:**
  - Naive Bayes is simple, fast and works well for text classification.
  - KNN is easy to implement and can model non-linear decision boundaries.
  - Both algorithms are effective for binary classification problems such as spam detection.
- **Limitations:**
  - Naive Bayes assumes feature independence, which may not always hold.
  - KNN has high prediction time because it computes distances to all training samples.
  - KNN is sensitive to the choice of k and requires feature scaling.

7 Experimental Setup

Python Version	3.13.13
Scikit-Learn Version	1.7.5
IDE	Jupyter Notebook
Operating System	Fedora 42
Hardware	Intel Core i5 (11th gen), 8GB RAM

8 Hyperparameter Selection

Parameter	Value
Number of Neighbors (k)	8
Distance Metric	Manhattan
Search Algorithm	KDTree and BallTree (Compared)
Weight Function	Distance
Cross Validation	5-Fold

9 Results

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Gaussian NB	0.8328	0.7146	0.9587	0.8188	0.9229
Multinomial NB	0.8936	0.9431	0.7769	0.8520	0.9625
Bernoulli NB	0.8795	0.8684	0.8182	0.8426	0.9457
Optimized KNN	0.9142	0.9356	0.8402	0.8853	0.9563

Table 2: Performance Comparison of Classification Models

10 Result Visualization

Confusion matrix optimized KNN

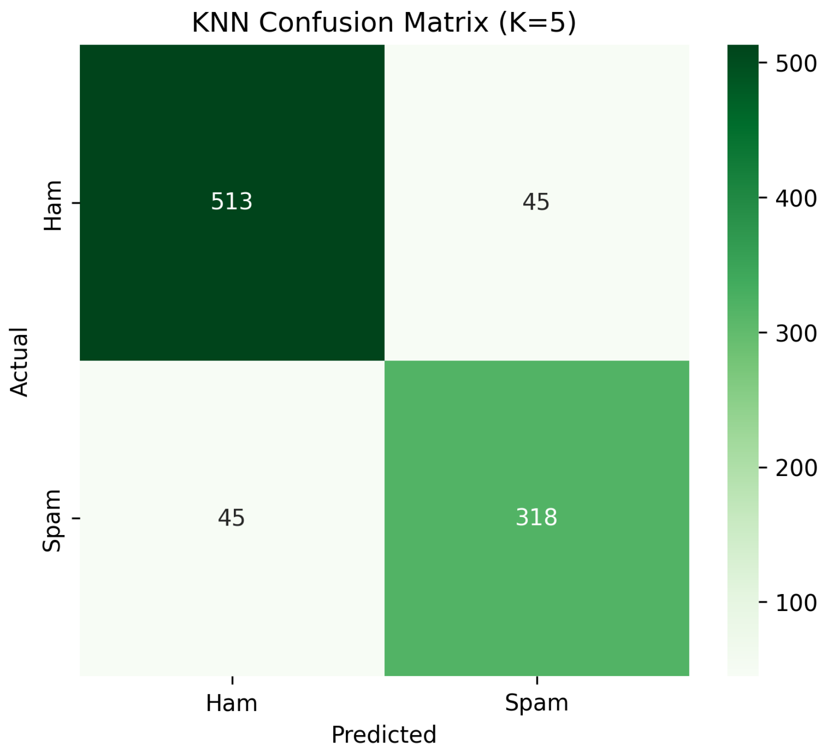

Figure 3: Confusion Matrix of Optimized KNN

Inference: The confusion matrix indicates that the optimized KNN classifier correctly classified most spam and ham emails with only a small number of misclassifications. The high number of true positives and true negatives demonstrates the effectiveness of the classifier. The low false positive and false negative counts indicate reliable prediction performance.

ROC Curve

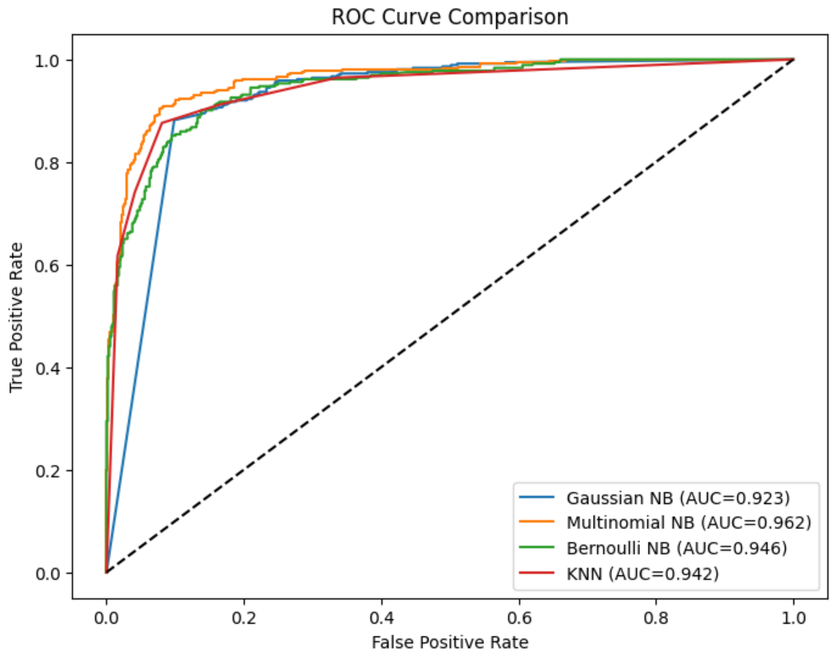

Figure 4: ROC Curve of the Optimized KNN Classifier

Inference:

The ROC curve illustrates the trade-off between the True Positive Rate (TPR) and False Positive Rate (FPR) at different classification thresholds. The curve lies close to the upper-left corner, indicating that the classifier effectively distinguishes spam emails from ham emails. A high Area Under the Curve (AUC) value demonstrates excellent classification performance and confirms the robustness of the optimized KNN model.

11 Discussion

The experiment compared three Naive Bayes variants with the KNN algorithm for email spam classification. Among the Naive Bayes classifiers, Multinomial Naive Bayes achieved the highest accuracy. Hyperparameter tuning using GridSearchCV further improved the performance of KNN, resulting in the highest overall accuracy. The comparison between KDTree and BallTree showed similar classification accuracy with differences only in execution time. K-Fold Cross Validation confirmed that the models produced consistent performance across different data splits.

12 Conclusion

The experiment successfully implemented Email Spam/Ham classification using Naive Bayes and KNN algorithms. Among the Naive Bayes variants, Multinomial Naive Bayes achieved the best performance. Hyperparameter optimization further improved KNN, making it the best-performing model with an accuracy of approximately 91.42%. The study demonstrated the effectiveness of hyperparameter tuning and cross-validation in improving model performance.

13 References

1. A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow*, 3rd ed., O'Reilly Media, 2022.

2. T. Hastie, R. Tibshirani and J. Friedman, *The Elements of Statistical Learning*. Springer, 2009.

3. Scikit-Learn Documentation. Available: https://scikit-learn.org

4. Kaggle Spambase Dataset. Available: https://www.kaggle.com/datasets/somesh24/spambase